CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

8.4 The Diffraction Grating

Syllabus Section: 8. Superposition **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (86 Marks)

Question 1 Easy [10 marks]

Part (a) [4 marks]

The diffraction grating equation is given by:

$$d \sin \theta = n\lambda$$

State the definition of the following variables and an appropriate unit for each.

(i) d

[1]

(ii) θ

[1]

(iii) n

[1]

(iv) λ

[1]

Mark Scheme (4 marks):

(i) *d* represents:

- Spacing between (adjacent) slits

  AND

- (Measured in) metres / m; *[1 mark]*

(ii) *θ* represents:

- Angular separation between (order of) maxima

  AND

- (Measured in) degrees / °; *[1 mark]*

(iii) *n* represents:

- Order of maxima

  AND

- It has no units; *[1 mark]*

(iv) *λ* represents:

- Wavelength (of the light source)

  AND

- (Measured in) metres / m; *[1 mark]*

[Total: 4 marks]

Part (b) [2 marks]

A diffraction grating has 720×10^3 lines per metres.

Calculate the distance between adjacent slits of the grating.

Mark Scheme (2 marks):

List the known quantities:

- Lines per m, $N = 720 \times 10^3$

Calculate the distance between adjacent slits, d:

- $d = \frac{1}{N}$
- $d = \frac{1}{720 \times 10^3}$ *[1 mark]*
- $d = 1.39 \times 10^{-6}$ m (1.39 μm) *[1 mark]*

[Total: 2 marks]

Part (c) [3 marks]

A narrow beam of light of wavelength 635 nm is incident normally on the diffraction grating, as shown in Fig. 1.1

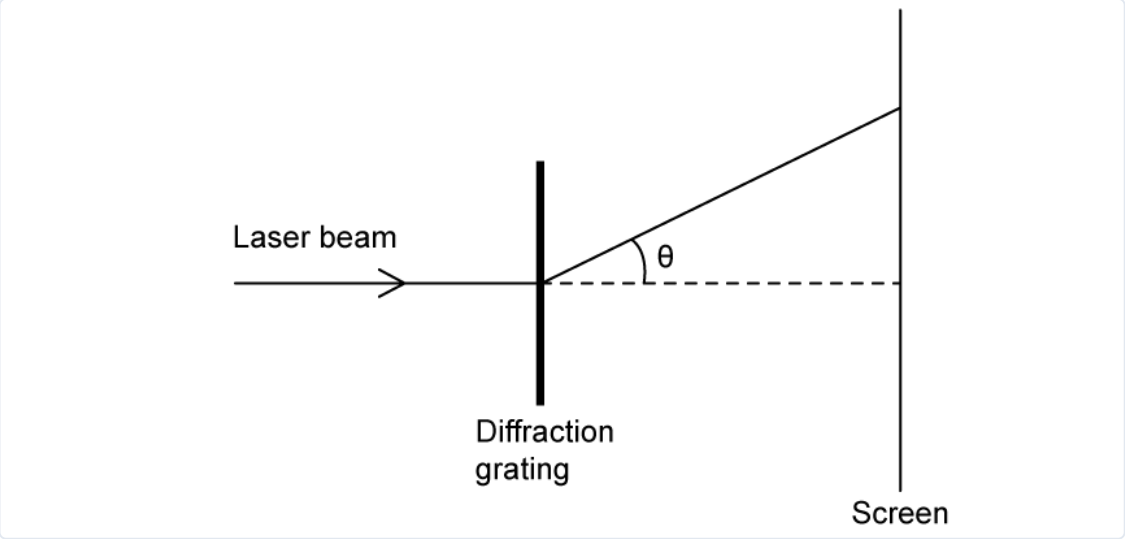


Fig. 1.1

The first-order diffraction maximum of the light is at an angle of θ to the direction of the incident light beam.

Calculate the value of θ.

Mark Scheme (3 marks):

Calculate the value of θ:

List the known quantities:

- Distance between adjacent slits, $d = 1.39 \times 10^{-6}$ m (from part b)
- Wavelength, $\lambda = 635$ nm $= 635 \times 10^{-9}$ m
- Order of maxima, $n = 1$

Rearrange the grating equation for θ:

- Grating equation: $d \sin \theta = n\lambda$
- Rearrange: $\sin \theta = \frac{n\lambda}{d}$
- $\theta = \sin^{-1} \frac{n\lambda}{d}$ *[1 mark]*

Substitute in values to calculate θ:

- $\theta = \sin^{-1} \frac{635 \times 10^{-9}}{1.39 \times 10^{-6}}$ *[1 mark]*
- $\theta = 27.2°$ *[1 mark]*

[Total: 3 marks]

*Remember to always have your calculator in **degrees** (and not radians) mode when calculating the angles using this equation!*

Part (d) **[1 mark]**

The highest order observed on the screen is the **second-order** diffraction maximum.

State how many minima are seen on the screen.

Mark Scheme (1 marks):

The number of minima seen on the screen is:

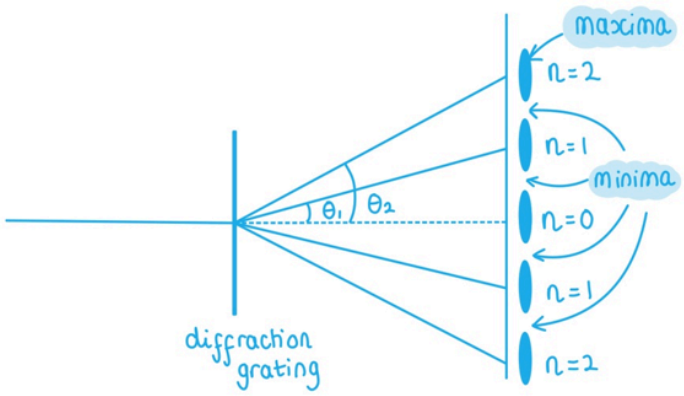
- 4 *[1 mark]*

  OR

- 6 (if extra minima counted at the ends) *[1 mark]*

[Total: 1 mark]

*These minima are easier to see from drawing a diagram. The minima are in between all the maxima. Remember that n starts from 0 and the subsequent orders are measured from **both** sides of the centre.*


maxima
n=2
n=1
minima
n=0
n=1
n=2
θ1
θ2
diffraction grating

Question 2 Easy [8 marks]

Part (a) [2 marks]

In a laboratory experiment, monochromatic light from a laser of wavelength λ is incident normally on a diffraction grating.

The diffracted laser light is projected on a white screen which is parallel to the plane of the grating. Fig 1.1 shows the position of the diffraction maxima.

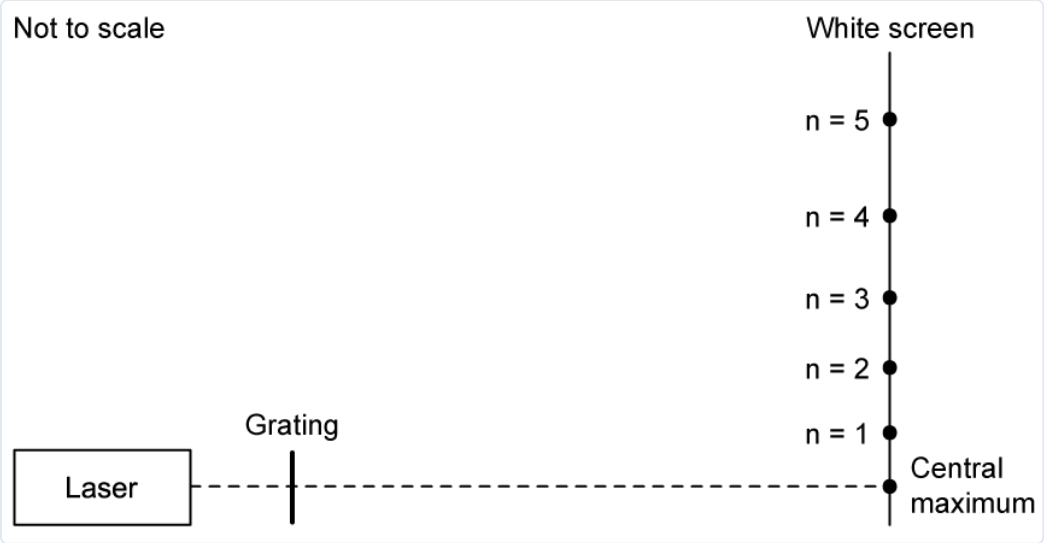
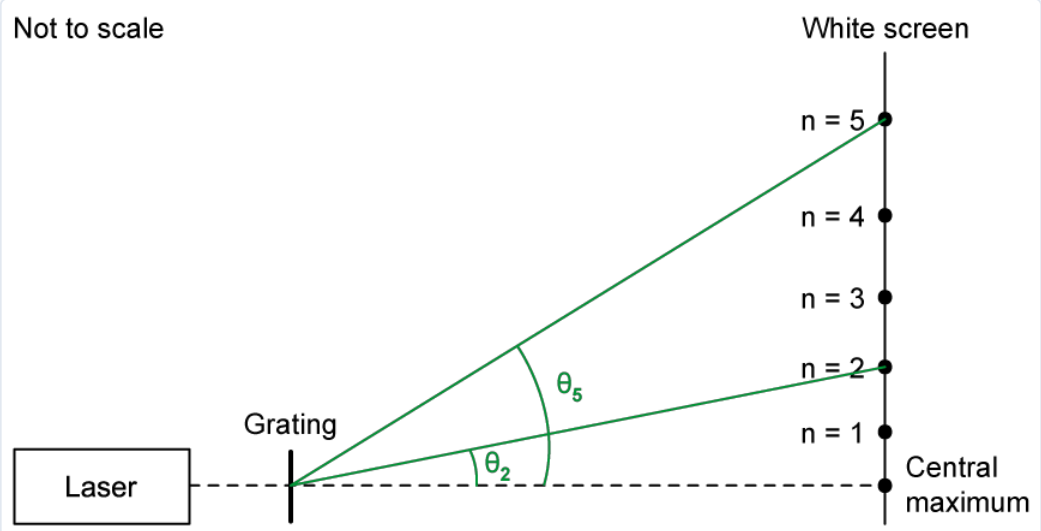


Fig. 1.1

On Fig. 1.1, sketch the diffracted laser light and mark the second and fifth-order diffraction maxima angles, θ_2 and θ_5 respectively.

Mark Scheme (2 marks):

The paths of the second and fifth-order diffraction maxima are as follows:



- Straight line from grating to n = 2 with angle labelled θ_2 from centre to line; *[1 mark]*
- Straight line from grating to n = 5 with angle labelled θ_5 from centre to line; *[1 mark]*

[Total: 2 marks]

Remember to always draw straight lines with a ruler or straight edge and clearly label your diagrams. If the sketches look too messy or vague, the examiner is unlikely to award you full marks. The angles should always be considered from the line connecting the grating to the central maximum (dashed line in Fig 1.1)

Part (b) **[2 marks]**

The spacing between each slit in the grating is 2.45 μm.

Calculate the number of lines per metre in this grating.

Mark Scheme (2 marks):

Calculate the number of lines per metre in this grating:

List the known quantities:

- Spacing between adjacent slits, d = 2.45 μm = 2.45 × 10^{-6} m

Calculate number of lines per metre, N:

- $N = \frac{1}{d}$
- $N = \frac{1}{2.45 \times 10^{-6}}$ *[1 mark]*
- N = 4.08 × 10^5 lines per metre *[1 mark]*

[Total: 2 marks]

Part (c) **[3 marks]**

The third–order spectral line is formed at an angle of 36º from the normal to the grating.

Calculate the value of λ.

Mark Scheme (3 marks):

List the known quantities:

- Spacing between adjacent slits, d = 2.45 μm = 2.45 × 10^{-6} m
- Order of maxima, n = 3
- Third-order diffraction angle, θ = 36°

Rearrange diffraction grating equation for λ:

- Grating equation: $d \sin \theta = n\lambda$ *[1 mark]*

- Rearrange: $\lambda = \frac{d \sin \theta}{n}$

Substitute in the values:

- $\lambda = \frac{2.45 \times 10^{-6} \sin 36^\circ}{3}$ *[1 mark]*
- λ = 480 × 10^{-9} m (480 nm) *[1 mark]*

[Total: 3 marks]

Part (d) **[1 mark]**

State the order that would produce the brightest image on the white screen.

Mark Scheme (1 marks):

The order that would produce the brightest image is:

- Zeroth order / *n* = 0 *[1 mark]*

[Total: 1 mark]

Question 3 Medium [14 marks]

Part (a) [3 marks]

A teacher explains a diffraction investigation to a group of physics students.

'In this investigation a laser will be used to shine monochromatic light of wavelength 581.9 nm onto a diffraction grating. The light passing through the grating will therefore be coherent.'

Define the following terms used by the teacher

(i) Monochromatic

[1]

(ii) Coherent

[2]

Mark Scheme (3 marks):

(i) Monochromatic is defined as:

- Having only one wavelength; *[1 mark]*

(ii) Coherent is defined as:

- Having the same frequency; *[1 mark]*
- And constant phase difference; *[1 mark]*

[Total: 3 marks]

Part (b) [6 marks]

During the investigation, a second-order maximum is produced at an angle of 35° measured from the normal to the grating.

(i) Calculate the number of lines per metre on the grating.

[3]

(ii) Calculate the highest order which is observable.

[3]

Mark Scheme (6 marks):

(i) Calculate the number of lines per metre on the grating:

List the known quantities:

- Wavelength, λ = 581.9 nm = 581.9 × 10^{-9} m

- Second order angle, θ = 35°
- Order, n = 2

Spacing between slits, d:

- $n\lambda = d \sin\theta$
- $d = \frac{n\lambda}{\sin\theta} = \frac{2 \times 581.9 \times 10^{-9}}{\sin 35}$ *[1 mark]*
- $d = 2.029 \times 10^{-6}$ m *[1 mark]*

Calculate the number of lines per metre:

- Number of lines per metre $N = \frac{1}{d} = \frac{1}{2.029 \times 10^{-6}}$
- $N = 4.93 \times 10^{5}$ m^{-1} *[1 mark]*

(ii) Calculate the highest order which is observable:

- The highest order would be at θ = 90° / $\sin\theta$ = 1 *[1 mark]*
- $n = \frac{d}{\lambda} = \frac{2.029 \times 10^{-6}}{581.9 \times 10^{-9}} = 3.49$ *[1 mark]*
- The highest order is the third / n = 3 *[1 mark]*

[Total: 6 marks]

Part (c) **[3 marks]**

Calculate the angular separation between the highest order and second order maxima.

Mark Scheme (3 marks):

Calculate the angular separation between the second and third maxima:

List the known quantities:

- Highest order, n = 3
- Second order angle, θ_2 = 35°
- Wavelength, λ = 581.9 nm

Calculate the third-order angle:

- $\sin \theta_3 = \frac{n\lambda}{d} = \frac{3 \times 581.9 \times 10^{-9}}{2.029 \times 10^{-6}}$ *[1 mark]*
- $\theta_3 = 59.36°$ *[1 mark]*

Calculate the angular separation:

- $\Delta\theta = \theta_3 - \theta_2$
- $\Delta\theta = 59 - 35 = 24°$ (2 s.f.) *[1 mark]*

[Total: 3 marks]

Part (d) [2 marks]

The investigation is repeated using the same grating but a different monochromatic source.

In this experiment, the second-order maximum is observed at an angle of 25.7°.

Calculate the wavelength of this second source.

Mark Scheme (2 marks):

Calculate the wavelength of this second source:

List the known quantities:

- Order, n = 2
- Second order angle, θ_2 = 25.7°
- Slit separation, d = 2.029 × 10^{-6} m

Calculate the wavelength:

- $n\lambda = d \sin\theta$
- $\lambda_{new} = \frac{d \sin\theta}{n} = \frac{2.029 \times 10^{-6} \times \sin 25.7}{2}$ *[1 mark]*
- $\lambda_{new} = 4.40 \times 10^{-7}$ m = 440 nm *[1 mark]*

[Total: 2 marks]

Question 4 Medium [11 marks]

Part (a) [3 marks]

A laser produces a monochromatic light of wavelength 581.9 nm which falls normally on a diffraction grating. A second order maximum is produced at an angle of 35° measured from the normal to the grating.

Calculate the number of lines per metre on the grating.

Mark Scheme (3 marks):

Calculate the number of lines per metre on the grating:

List the known quantities:

- Wavelength, λ = 581.9 nm
- Second order angle, θ = 35°
- Order, n = 2

Calculate the spacing between slits:

- Grating equation: $d \sin \theta = n\lambda \Rightarrow d = \frac{n\lambda}{\sin \theta}$
- Slit separation: $d = \frac{2 \times 581.9 \times 10^{-9}}{\sin 35}$ *[1 mark]*
- Slit separation: d = 2.029 × 10^{-6} m *[1 mark]*

Calculate the number of lines per metre:

- Number of lines per metre: $N = \frac{1}{d}$
- Number of lines per metre: $N = \frac{1}{2.029 \times 10^{-6}}$ = 4.93 × 10^{5} m^{-1} *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

Calculate the highest order which is observable.

Mark Scheme (3 marks):

Calculate the highest order which is observable:

List the known quantities:

- Slit separation, d = 2.029 × 10^{-6} m (from (a))
- Wavelength, λ = 581.9 nm = 581.9 × 10^{-9} m

The highest order would be observed at:

- Diffraction angle: θ = 90° / $\sin \theta$ = 1 *[1 mark]*

- Grating equation: $d \sin \theta = n\lambda \Rightarrow n = \frac{d \sin 90°}{\lambda} = \frac{d}{\lambda}$
- Highest order: $n = \frac{2.029 \times 10^{-6}}{581.9 \times 10^{-9}}$ = 3.49 *[1 mark]*
- The highest order is the third / n = 3 *[1 mark]*

[Total: 3 marks]

*Make sure to always **round down** when calculating diffraction orders*

Part (c) **[3 marks]**

Calculate the angular separation between the highest order and second order maxima.

Mark Scheme (3 marks):

Calculate the angular separation between the n = 2 and n = 3:

List the known quantities:

- Wavelength, λ = 581.9 nm = 581.9 × 10^{-9} m
- Slit separation, d = 2.029 × 10^{-6} m (from (a))
- Second order, n = 2
- Highest order, n = 3
- Second-order angle, θ_2 = 35°

Calculate the third-order diffraction angle:

- Grating equation: $d \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n\lambda}{d}$
- Third-order angle: $\sin \theta_3 = \frac{3\lambda}{d} = \frac{3 \times 581.9 \times 10^{-9}}{2.029 \times 10^{-6}}$ *[1 mark]*
- Third-order angle: $\theta_3 = \sin^{-1} \frac{3 \times 581.9 \times 10^{-9}}{2.029 \times 10^{-6}}$ = 59.4° *[1 mark]*

Calculate the angular separation:

- Angular separation: $\Delta\theta = \theta_3 - \theta_2$
- Angular separation: $\Delta\theta$ = 59 – 35 = 24° *[1 mark]*

[Total: 3 marks]

Don't worry if you got the answers incorrect in parts (a) and (b), whatever answers you obtained, as long as you used them correctly here, you would still gain full marks - this is known as 'e.c.f.' or 'error carried forward' marks

Part (d) **[2 marks]**

When the same grating is used with a different monochromatic source, the second-order maximum is observed at an angle of 25.7°.

Calculate the wavelength of this second source.

Mark Scheme (2 marks):

Calculate the wavelength of the second source:

List the known quantities:

- Diffraction order, n = 2
- Second order angle, θ_2 = 25.7°
- Slit separation, d = 2.029 × 10^{-6} m (from (a))

Calculate the new wavelength:

- Grating equation: $n\lambda = d \sin \theta \Rightarrow \lambda = \frac{d \sin \theta}{n}$
- New wavelength: $\lambda_{new} = \frac{2.029 \times 10^{-6} \times \sin 25.7°}{2}$ *[1 mark]*
- New wavelength: λ_{new} = 4.40 × 10^{-7} m = 440 nm *[1 mark]*

[Total: 2 marks]

Question 5 Medium [11 marks]

Part (a) [4 marks]

A helium-neon laser produces monochromatic light of wavelength 672 nm which falls perpendicular on a diffraction grating which has 350 lines per mm.

Calculate the angle between the second-order maxima.

Mark Scheme (4 marks):

Calculate the angle between the second-order maxima:

List the known quantities:

- Number of lines, N = 350 lines per mm = 350 × 10^3 lines per metre
- Wavelength, λ = 672 nm = 672 × 10^{-9} m

Calculate the spacing between the slits:

- Slit separation: $d = \frac{1}{N}$
- Slit separation: $d = \frac{1}{350 \times 10^3}$ = 2.86 × 10^{-6} m *[1 mark]*

Calculate the angle between the central and second-order maxima:

- Grating equation: $d \sin\theta = n\lambda \Rightarrow \sin\theta = \frac{n\lambda}{d}$
- Diffraction angle: $\sin\theta = \frac{2 \times 672 \times 10^{-9}}{2.86 \times 10^{-6}}$ *[1 mark]*
- Diffraction angle: $\theta = \sin^{-1}\frac{2 \times 672 \times 10^{-9}}{2.86 \times 10^{-6}}$ = 28° *[1 mark]*

Determine the angle between both second-order maxima:

- Angle between both second-order maxima = 2 × 28° = 56° *[1 mark]*

[Total: 4 marks]

*The angle we calculated in the second step is from the central maxima to **one** of the second-order maxima, remember, there are maxima on each side of the central maxima, so the final mark is for recognising this and multiplying your angle by 2*

Part (b) [3 marks]

Show that the highest order image shown in this arrangement is n = 4.

Mark Scheme (3 marks):

To show that the highest order image shown in this arrangement is n = 4:

- The highest order would be at θ = 90° / $\sin\theta$ = 1 *[1 mark]*
- $n = \frac{d}{\lambda} = \frac{2.86 \times 10^{-6}}{672 \times 10^{-9}}$ = 4.26 *[1 mark]*

- The highest order is the fourth / n = 4

OR

Substitute in *n = 5:*

- $\sin\theta = \frac{n\lambda}{d} = \frac{5 \times 672 \times 10^{-9}}{2.86 \times 10^{-6}}$ = 1.17 *[1 mark]*
- $\sin\theta$ cannot be greater than 1 *[1 mark]*

Final statement:

- Therefore, the fourth order is the maximum order possible / visible *[1 mark]*

[Total: 3 marks]

Part (c) [2 marks]

A different laser that produces monochromatic light of wavelength λ falls perpendicular to a diffraction grating from which adjacent lines are 5λ apart.

Calculate the angle for the third order maximum.

Mark Scheme (2 marks):

To calculate the angle for the third order maximum:

List the known quantities:

- $d = 5\lambda$

Calculate the angle of third order maximum:

- $\sin\theta_3 = \frac{3\lambda}{5\lambda} = \frac{3}{5}$ *[1 mark]*
- $\theta_3 = \sin^{-1}\frac{3}{5} = 37°$ *[1 mark]*

[Total: 2 marks]

Part (d) [2 marks]

The diffraction grating is placed 2.7 m from a screen.

Hence or otherwise, calculate the distance between the third diffraction order and the normal to the grating as measured on the screen.

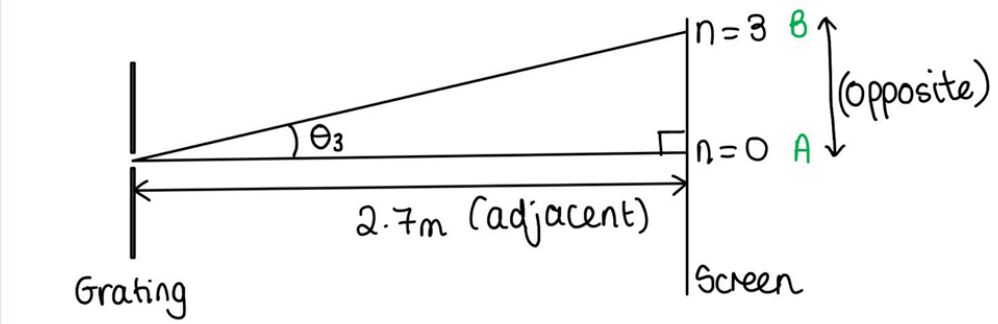
Mark Scheme (2 marks):

To calculate the distance between the third diffraction order and the normal to the grating:

List the known quantities:

- Angle to third order maximum, $\theta_3 = 37°$ (from part c)

- Distance from grating to screen = 2.7 m



Calculate from centre to third order:

- **AB** = 2.7 × tan (37) *[1 mark]*
- **AB** = 2.0 m *[1 mark]*

[Total: 2 marks]

Drawing out a diagram here to see which sides are the adjacent, hypotenuse and opposite of the right-angled triangle will make this question to visualise.

Question 6 Hard [15 marks]

Part (a) [5 marks]

Students in a laboratory have created the following set-up shown in Fig 1.1. Parallel rays of monochromatic light from two adjacent slits A and B of wavelength λ are incident normally on a diffraction grating with a slit separation d.

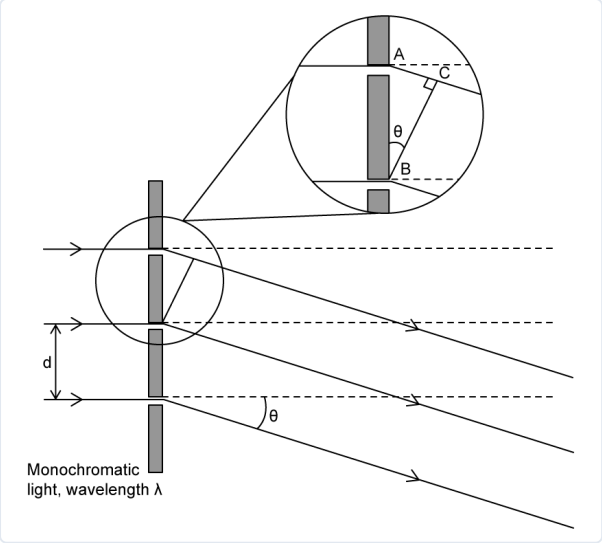


Fig. 1.1

Use the diagram to derive the equation $n\lambda = d\sin\theta$ where θ is the angle of diffraction of a maxima order n visible on a screen.

Mark Scheme (5 marks):

Consider the path difference in terms of wavelength, λ when bright fringes are created:

- Constructive interference between two parallel rays of light from two adjacent slits A and B results in bright fringes of light known as maxima; *[1 mark]*
- AC is the path difference between the two parallel rays of light; *[1 mark]*
- Constructive interference occurs when the path difference between the two rays is a whole number of wavelengths
    - Therefore, AC = $n\lambda$, where n is an integer and corresponds to the order of maxima; *[1 mark]*

Consider the path difference in terms of trigonometry:

- Using trigonometry:
    - $sin\theta = \frac{AC}{AB}$
    - So, AC = $d \times \sin\theta$ *[1 mark]*

Equate the two expressions for path difference, AC:

- Therefore the equation to calculate the angle of diffraction, θ of each order of maxima n is:

- $n\lambda = d\sin\theta$ as required; *[1 mark]*

[Total: 5 marks]

Part (b) [5 marks]

The monochromatic light in the set-up in part (a) has a wavelength of 499 nm. The graph in Fig. 1.2 shows the variation of $\sin\theta$ with the order n of the maximum. The central order corresponds to $n = 0$.

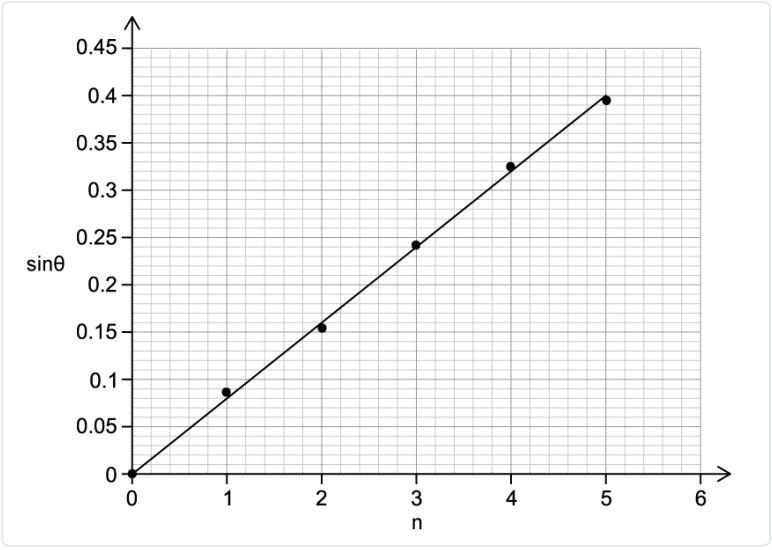


Fig. 1.2

Determine a mean value for the number of slits per mm of the grating.

Mark Scheme (5 marks):

List the known quantities:

- Wavelength of light, λ = 499 nm = 499 × 10^{-9} m

Obtain an expression for the gradient of the graph:

- The gradient of a straight line, $\frac{\Delta y}{\Delta x}$
- Therefore, $\frac{\Delta \sin\theta}{\Delta n} = \frac{\sin\theta}{n}$ *[1 mark]*

Consider the gradient of the graph in terms of the diffraction grating equation:

- Recall: $n\lambda = d\sin\theta$
- Dividing by d and n gives, $\frac{\lambda}{d} = \frac{\sin\theta}{n}$ *[1 mark]*

Use the graph to calculate the value of the gradient:

- $\Delta\sin\theta$ = 0.39
- Δn = 5
- So, gradient = $\frac{0.39}{5}$ = 0.078 *[1 mark]*

Consider the gradient in terms of *d*:

- Gradient = $\frac{\lambda}{d}$ so $d = \frac{\lambda}{Gradient}$
- $d = \frac{499 \times 10^{-9}}{0.078}$
- $d = 6.40 \times 10^{-6}$ m *[1 mark]*

Convert the slit separation to mm and calculate the mean number of slits per mm, *N*:

- 1000 mm = 1 m
- Therefore, $6.40 \times 10^{-6} \times 1000 = 6.40 \times 10^{-3}$ mm
- Hence, in 1 mm there are $\frac{1}{6.40 \times 10^{-3}}$ slits
- Therefore, *N* = 156.25 = 156 *[1 mark]*

[Total: 5 marks]

Part (c) **[3 marks]**

The grating is 30 mm wide.

Determine the number of slits required to obtain a maximum of five bright fringes on the screen.

Mark Scheme (3 marks):

List the known quantities:

- Wavelength, λ = 499 nm = 499×10^{-9} m (from part (b))
- Grating width = 30 mm = 30×10^{-3} m
- Number of maxima, *n* = 2

Consider the angle of diffraction required to obtain a maximum of 5 bright fringes:

- The maximum angle of diffraction is 90° *[1 mark]*

Rearrange the diffraction grating equation and substitute in the known quantities to calculate the slit separation, *d*:

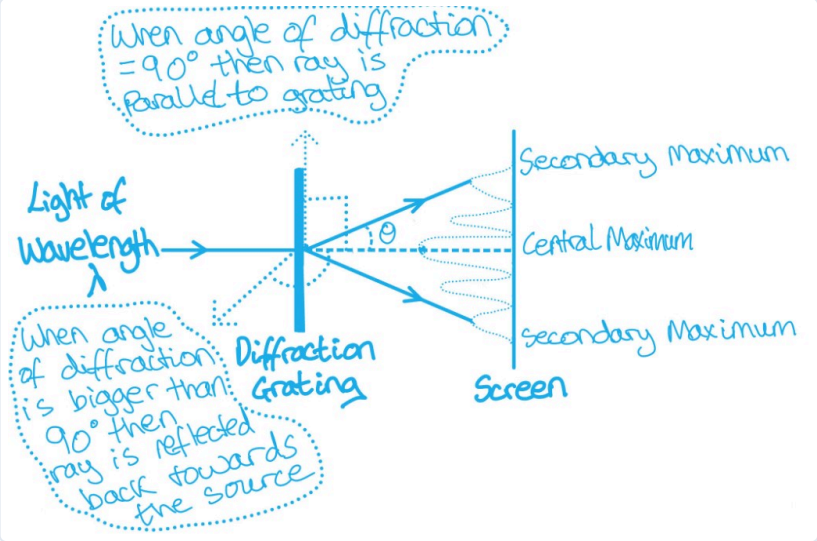
- Recall: $n\lambda = d\sin\theta$
- $d = \frac{n\lambda}{\sin\theta}$
- $d = \frac{2 \times (499 \times 10^{-9})}{\sin(90)}$
- $d = 9.98 \times 10^{-7}$ m *[1 mark]*

Calculate the number of slits required in the grating 40 mm wide:

- The grating is 30×10^{-3} m wide, so there are $\frac{30 \times 10^{-3}}{9.98 \times 10^{-7}}$ slits
- So the number of slits = 30060 *[1 mark]*

[Total: 3 marks]

Recognising that the maximum angle of diffraction is 90° and using n = 5 at this angle is the key to answering this question correctly. The diagram shows why 90° is the maximum angle of diffraction.



Part (d) [2 marks]

The students claim that they observed the following diffraction pattern in Fig. 1.3 on the screen for the grating from part (c).

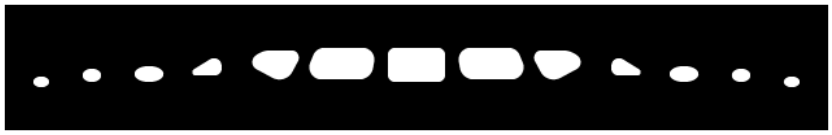


Fig. 1.3

State two reasons why the interference pattern obtained cannot be correct.

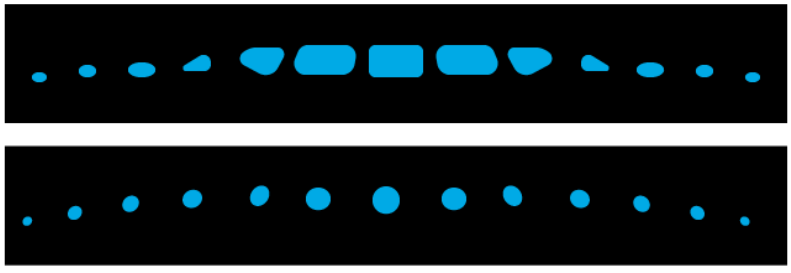
Mark Scheme (2 marks):

The diffraction pattern obtained cannot be correct because:

- The bright fringes are too wide; *[1 mark]*
- The dark fringes are too narrow (for a diffraction grating containing 30060 slits); *[1 mark]*

[Total: 2 marks]

It can be difficult to identify problems with diffraction patterns, especially isolated patterns but putting it together with another diffraction pattern makes it possible to compare them.



Double slit diffraction pattern

Diffraction grating diffraction pattern

Question 7 Hard [8 marks]

Part (a) [3 marks]

A narrow beam of coherent light of wavelength 520 nm is incident upon a diffraction grating that has 3.10 × 10^2 lines per mm. The diffraction pattern is seen on a screen.

Calculate the maximum number of minima that could be seen on the screen.

Mark Scheme (3 marks):

List the known quantities:

- Wavelength, λ = 520 nm = 520 × 10^{-9} m
- N = 3.10 × 10^2 lines per mm = 3.10 × 10^5 lines per m

Calculate adjacent spacing between slits, d:

- $d = \frac{1}{N} = \frac{1}{3.10 \times 10^5}$ = 3.23 × 10^{-6} m *[1 mark]*

Calculate the highest order possible:

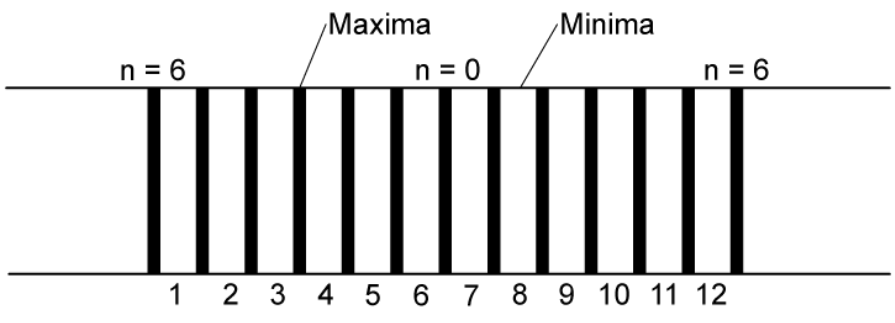
- Recall: $d \sin\theta = n\lambda$
- The highest order would be at θ = 90° **OR** $\sin\theta$ = 1
- So $d \sin(90) = n\lambda$ becomes $d = n\lambda$
- $n = \frac{d}{\lambda} = \frac{3.23 \times 10^{-6}}{520 \times 10^{-9}}$ = 6.21
- The highest order is the sixth (n = 6) *[1 mark]*

Total minima:

- Total minima = (2 × 6) = 12 *[1 mark]*

[Total: 3 marks]

Remember, for a grating, there is one minimum between each pair of adjacent maxima. With maxima from n = -6 to n = +6 plus the central maximum, there are 13 maxima and therefore 12 minima. There are no additional minima beyond the outermost maxima.



Part (b) [2 marks]

A different light source is incident upon the same diffraction grating. A student suspects there are two wavelengths of light in this incident beam, one of wavelength 490 nm and another of wavelength 490.2 nm.

Determine the highest order of the visible diffracted light.

Mark Scheme (2 marks):

List the known quantities:

- d = 3.23 × 10^{-6} m (from part a)
- Wavelength,λ = 490 nm = 490 × 10^{-9} m

Explain where the different wavelengths of light are most distinguishable:

- They are most likely to be distinguishable at the highest order; *[1 mark]*

Calculate the highest order visible:

- The highest order would be at θ = 90° **OR** $\sin \theta = 1$
- $n = \frac{d}{\lambda} = \frac{3.23 \times 10^{-6}}{490 \times 10^{-9}}$ = 6.59 *[1 mark]*
- The highest order is the sixth (n = 6)

[Total: 2 marks]

*Remember to always round **down** for the value of your highest order since n indicates the number of visible maxima. For example, n = 4.8 means there will be at most 4 visible maxima (one central maxima, and three maxima either side of this). Note, we cannot round up to 5, because there are not 5 maxima visibly created.*

Part (c) [2 marks]

The minimum angular separation for which the two wavelengths may be differentiated is 0.05°.

Deduce whether the student observed the two wavelengths as distinct images at this order. Support your answer with calculations ignoring the blurring effect often observed at higher order maxima.

Mark Scheme (2 marks):

List the known quantities:

- Wavelength, λ_1 = 490 nm = 490 × 10^{-9} m
- Wavelength, λ_2 = 490.2 nm = 490.2 × 10^{-9} m
- Slit spacing, d = 3.23 × 10^{-6} m (from part a)
- Highest order visible n = 6 (from part b)

Calculate the angle for the highest order visible for the first wavelength, θ_1:

- $n\lambda = d \sin\theta$ therefore $\theta = \sin^{-1} \frac{n\lambda}{d}$

- $\theta_1 = \sin^{-1} \frac{6 \times 490 \times 10^{-9}}{3.23 \times 10^{-6}} = 65.535^\circ$

Calculate the angle of the highest order visible for the second wavelength, θ_2:

- $\theta_2 = \sin^{-1} \frac{6 \times 490.2 \times 10^{-9}}{3.23 \times 10^{-6}} = 65.587^\circ$

Calculate the difference in angles:

- $\Delta\theta = \theta_2 - \theta_1 = 65.587^\circ - 65.535^\circ = 0.052^\circ$ *[1 mark]*

Conclusion

- Since $0.052^\circ > 0.05^\circ$ the student can observe the two wavelengths as separate distinct images; *[1 mark]*

[Total: 2 marks]

Part (d) **[1 mark]**

Fig. 1.1 shows the shortest wavelength of this light incident on the screen after the other has been removed from part c).

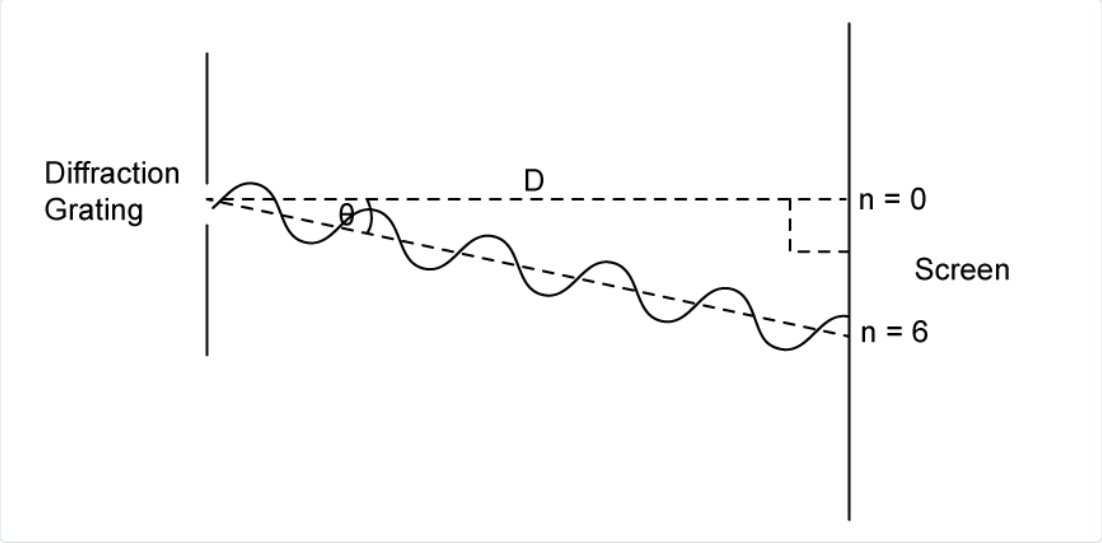


Fig. 1.1

Determine the distance between the diffraction grating and the screen, *D*.

Mark Scheme (1 marks):

List the known quantities:

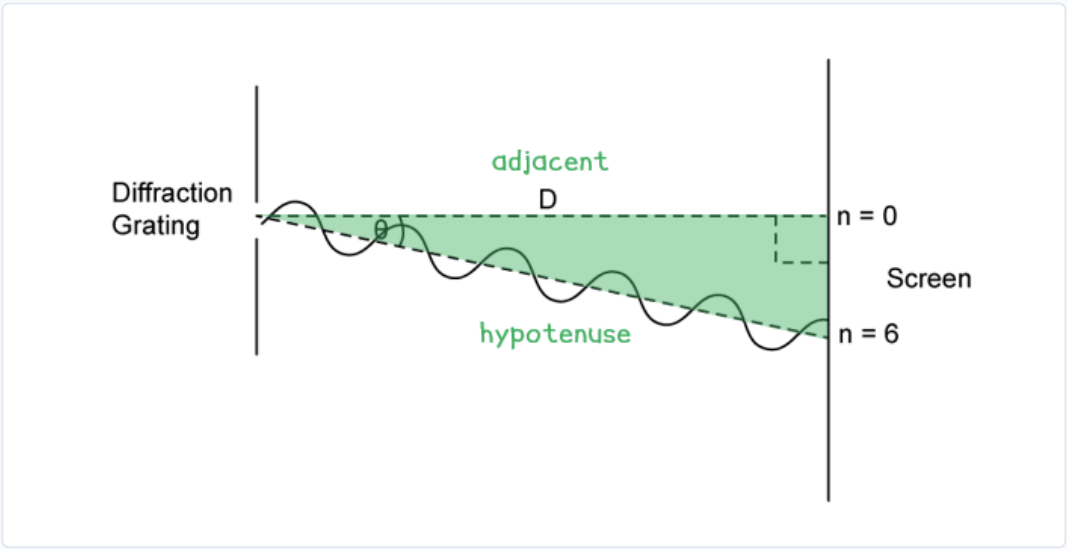
- Wavelength of light, λ = 490 nm = 490 × 10^{-9} m (from part b)
- Angle of diffraction, θ = 65.535° (from part c)

Determine the distance for the hypotenuse, *H* of the triangle from slit to sixth order:

- The number of complete wavelengths between the grating and the screen = 5.25 (from Fig. 1.1)

- Hence, the distance is 5.25λ
- $H = 5.25 \times (490 \times 10^{-9}) = 2.47 \times 10^{-6}$ m *[1 mark]*

Use trigonometry to find the length of the adjacent of the triangle, *D*:



- adj = cosθ × hyp
- $D = \cos\theta \times H$
- $D = \cos(65.535) \times (2.47 \times 10^{-6})$
- $D = 1.07 \times 10^{-6}$ m *[1 mark]*

[Total: 2 marks]

Look at the information on this diagram carefully. You can see that there are five and a quarter wavelengths of light between the slit and the screen, so this can give you the length of the hypotenuse of the triangle.

Make sure you obtain the correct information from the previous parts of this question too.

Drawing the triangle is essential, so you do not make a mistake with your trigonometry.

Question 8 Hard [9 marks]

Part (a) [2 marks]

Monochromatic light is incident normally on a diffraction grating.

Describe the diffraction of the light waves at the grating.

Mark Scheme (2 marks):

When light is incident on the diffraction grating:

- The light wave(s) pass through / enter the slit(s) *[1 mark]*
- And spread (into a geometric shadow) *[1 mark]*

Part (b) [4 marks]

A diffraction pattern forms on a screen.

The light has a wavelength of 630 nm. The diffraction grating has 4.50×10^2 lines per millimetre.

Determine the total number of gaps between orders, formed on the screen.

Mark Scheme (4 marks):

To determine the total number ofgaps between orders, formed on the screen:

- List the known quantities:
  - Wavelength, λ = 630 nm = 630×10^{-9} m
  - Number of lines, N = 4.50×10^2 lines mm^{-1}
- Recall the diffraction grating equation:

$$d \sin \theta = n\lambda \text{ } [1 \text{ mark}]$$

- Calculate the separation d of the lines in the grating in metres:

$$N = \frac{4.50 \times 10^2}{10^{-3}} = 4.50 \times 10^5 \text{ lines } m^{-1}$$

$$d = \frac{1}{N} = \frac{1}{4.50 \times 10^5}$$

$$d = 2.22 \times 10^{-6} \text{ m } [1 \text{ mark}]$$

- Determine the highest diffraction order n produced on the screen:

The highest possible order is at $\theta = 90°$ **OR** $\sin \theta = 1$

$$d \sin 90° = n\lambda$$

$$n = \frac{d}{\lambda} \text{ } [1 \text{ mark}]$$

$$n = \frac{2.22 \times 10^{-6}}{630 \times 10^{-9}}$$

$$n = 3.52...$$

The highest order produced is $n = 3$

- Determine the total number of gaps formed on the screen:

Total number of gaps = 2 × 3

Total number of gaps = *6 [1 mark]*

1 mark for stating the diffraction grating equation $d \sin \theta = n\lambda$

1 mark for $d = \frac{1}{4.50 \times 10^5}$ **OR** $d = 2.22 \times 10^{-6}$ m

1 mark for $d \sin 90° = n\lambda$ **OR** $d = n\lambda$ **OR** $n = \frac{d}{\lambda}$

1 mark for the total number of gaps = $2n$ **AND** given as an integer

Remember to always round your calculated value of n down rather than up. This is because n indicates the total number of visible maxima.

In a diffraction pattern, there is one gap between each pair of adjacent maxima. In this case, there are three diffraction maxima on either side of the central maximum, and therefore 6 gaps between these as visible on the screen.

Part (c) **[3 marks]**

The beam of light in **(b)** is replaced by a beam of blue light incident on the same diffraction grating.

(i) State whether the frequency of the blue light is greater than, less than, or the same as the frequency of the original light.

[1]

(ii) State and explain the effect of this change on the number of gaps formed on the screen. A calculation is not required.

[2]

Mark Scheme (3 marks):

(i) The frequency of the blue light is

- Greater than (the frequency of the original light) *[1 mark]*

(ii) The effect of this change on the number of gaps formed is:

- The number (of gaps formed) is greater **OR** more gaps form (on the screen) *[1 mark]*

Explanation:

- The wavelength of blue light is shorter (than 630 nm / wavelength of original light) so the maximum order n_{max} increases $n = \frac{d}{\lambda}$ and more diffraction orders appear on the screen *[1 mark]*